In [ ]:

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import tkinter as tk
from matplotlib.backends.backend_tkagg import FigureCanvasTkAgg
import matplotlib.pyplot as plt
import numpy as np

def tent_map(mu, x):
    return mu * x if x < 0.5 else mu * (1 - x)

def generate_iteration(mu=2.0, x0=0.1, n=100):
    x_vals = [x0]
    x = x0
    for _ in range(n):
        x = tent_map(mu, x)
        x_vals.append(x)
    return x_vals

def generate_poincare(mu=2.0, x0=0.1, n=500):
    full_x_vals = generate_iteration(mu, x0, n + 100)
    return full_x_vals[:-1], full_x_vals[1:]

def generate_bifurcation(mu_min=0, mu_max=2, x0=0.1, n_transient=1000, n_iter=100):
    mu_vals = np.linspace(mu_min, mu_max, 1000)
    xs = []
    mus = []

    for mu in mu_vals:
        x = x0
        for _ in range(n_transient):
            x = tent_map(mu, x)
        for _ in range(n_iter):
            x = tent_map(mu, x)
            mus.append(mu)
            xs.append(x)

    return mus, xs

canvas_widget = None

def show_plot(fig):
    global canvas_widget

    if canvas_widget:
        canvas_widget.destroy()

    canvas = FigureCanvasTkAgg(fig, master=plot_frame)
    canvas_widget = canvas.get_tk_widget()

    canvas_widget.pack(side=tk.TOP, fill=tk.BOTH, expand=1)

    canvas.draw()

    plt.close(fig)

def plot_iteration():
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    mu_val = 2
    x0_val = 0.1
    n_iterations = 100

    x_vals = generate_iteration(mu=mu_val, x0=x0_val, n=n_iterations)

    fig = plt.Figure(figsize=(6, 4), dpi=100)
    ax = fig.add_subplot(111)

    ax.plot(range(len(x_vals)), x_vals, 'o-', markersize=3, linewidth=0.8, color='b

    ax.set_title(f"Iteration Plot (Tent Map, μ={mu_val})")
    ax.set_xlabel("Iteration (n)")
    ax.set_ylabel("x_n")

    ax.grid(True, linestyle='--', alpha=0.6)

    show_plot(fig)

def plot_poincare():
    mu_val = 2.0
    x0_val = 0.1
    n_points = 500

    x_n_values, x_n_plus_1_values = generate_poincare(mu=mu_val, x0=x0_val, n=n_poi

    fig = plt.Figure(figsize=(6, 4), dpi=100)
    ax = fig.add_subplot(111)

    ax.plot(x_n_values, x_n_plus_1_values, '.r', markersize=3, alpha=0.7)

    ax.set_title(f"Poincaré Plot (Tent Map, μ={mu_val})")
    ax.set_xlabel("x_n")
    ax.set_ylabel("x_n+1")

    ax.grid(True, linestyle='--', alpha=0.6)

    ax.set_xlim(0, 1)
    ax.set_ylim(0, 1)

    show_plot(fig)

def plot_bifurcation():
    mu_min_val = 0.0
    mu_max_val = 2.0
    x0_val = 0.1
    n_transient_val = 1000
    n_iter_val = 100

    mus, xs = generate_bifurcation(mu_min=mu_min_val, mu_max=mu_max_val,
                                   x0=x0_val, n_transient=n_transient_val,
                                   n_iter=n_iter_val)

    fig = plt.Figure(figsize=(6, 4), dpi=100)
    ax = fig.add_subplot(111)
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    ax.plot(mus, xs, ',k', alpha=0.25)

    ax.set_title("Bifurcation Diagram (Tent Map)")
    ax.set_xlabel("μ (mu)")
    ax.set_ylabel("x")

    ax.grid(True, linestyle='--', alpha=0.6)

    ax.set_ylim(0, 1)

    show_plot(fig)

root = tk.Tk()
root.title("Tent Map Analysis")
root.geometry("700x550")

button_frame = tk.Frame(root, pady=10)
button_frame.pack(side=tk.TOP, fill=tk.X)

tk.Button(button_frame, text="Iteration Plot", command=plot_iteration).pack(side=tk
tk.Button(button_frame, text="Poincaré Plot", command=plot_poincare).pack(side=tk.L
tk.Button(button_frame, text="Bifurcation Diagram", command=plot_bifurcation).pack(

plot_frame = tk.Frame(root, bg="white", width=600, height=400, bd=2, relief="groove
plot_frame.pack(side=tk.BOTTOM, fill=tk.BOTH, expand=True, padx=10, pady=10)
plot_frame.pack_propagate(False)

root.mainloop()
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In [ ]:
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